Integral Inequalities

From Basic Bounds to Orthogonal Projections

April 2026

Core Properties of Definite Integrals

Monotonicity. If $f(x) \leq g(x)$ for all $x \in [a, b]$, then

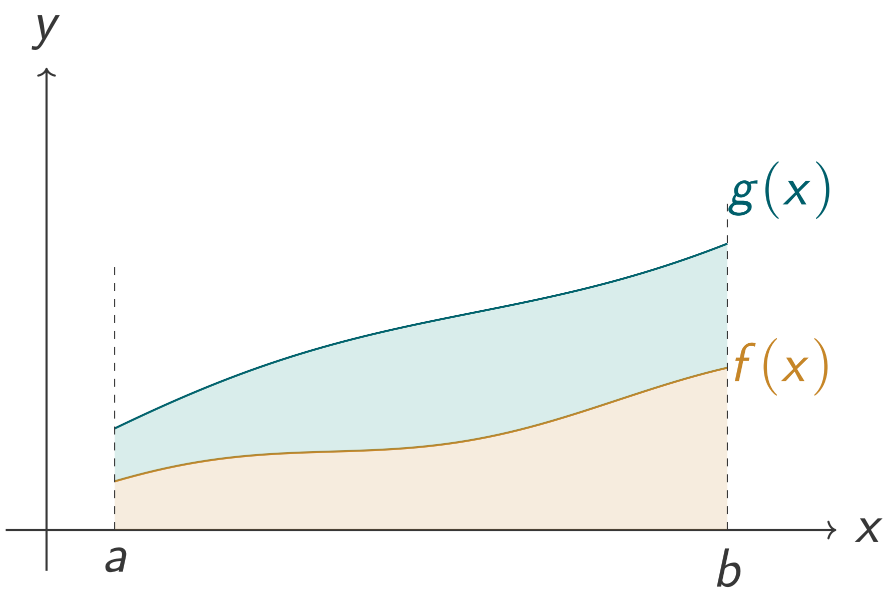
$$\int_a^b f(x)\,dx \;\leq\; \int_a^b g(x)\,dx.$$

Triangle inequality.

$$\left|\int_a^b f(x)\,dx\right| \;\leq\; \int_a^b |f(x)|\,dx.$$

Both are direct consequences of the definition of the integral as a limit of Riemann sums: inequalities that hold pointwise survive the limit.

Geometric Picture: Monotonicity



The area under g dominates the area under f on $[a, b]$.

Example 1 — Problem

Problem

Prove that $0 \leq \int_0^1 x^n \sin(\pi x)\, dx \leq \dfrac{1}{n+1}$ for $n \in \mathbb{N}$.

The integrand is a product of two familiar pieces on $[0, 1]$: the power x^n and the sine curve $\sin(\pi x)$. Both are non-negative here, so the integral is non-negative. The upper bound should come from replacing the harder factor by its largest value on the interval.

Example 1 — Proof

For $x \in [0,1]$ the argument πx lies in $[0,\pi]$, on which the sine function is non-negative and bounded by 1:

$$0 \le \sin(\pi x) \le 1.$$

Multiplying by the non-negative quantity x^n,

$$0 \;\le\; x^n \sin(\pi x) \;\le\; x^n.$$

Integrating over $[0,1]$ and using monotonicity,

$$0 \;\le\; \int_0^1 x^n \sin(\pi x)\, dx \;\le\; \int_0^1 x^n\, dx \;=\; \frac{1}{n+1}. \quad \square$$

Example 2 — Problem

Problem

Determine strict bounds for $I = \int_0^1 e^{-x^2}\,dx$.

The integrand e^{-x^2} has no elementary antiderivative, so we bracket it between two simpler functions whose integrals we can compute exactly. The natural candidates come from the exponential inequality $1 + y \leq e^y$.

Example 2 — Proof

The inequality $1 + y \le e^y$ holds for all real y, with equality only at $y = 0$. Setting $y = -x^2$,

$$1 - x^2 \;\le\; e^{-x^2}, \qquad \text{strict for } x \neq 0.$$

Moreover, for $x \in (0,1)$ we have $-x^2 < 0$, so $e^{-x^2} < 1$. Thus on $(0,1)$,

$$1 - x^2 \;<\; e^{-x^2} \;<\; 1.$$

Integrating over $[0,1]$,

$$\int_0^1 (1 - x^2)\, dx \;<\; \int_0^1 e^{-x^2}\, dx \;<\; \int_0^1 1\, dx \implies \frac{2}{3} \;<\; I \;<\; 1. \quad \square$$

Cauchy–Schwarz Inequality

Theorem

Discrete form. For real numbers a_i, b_i,

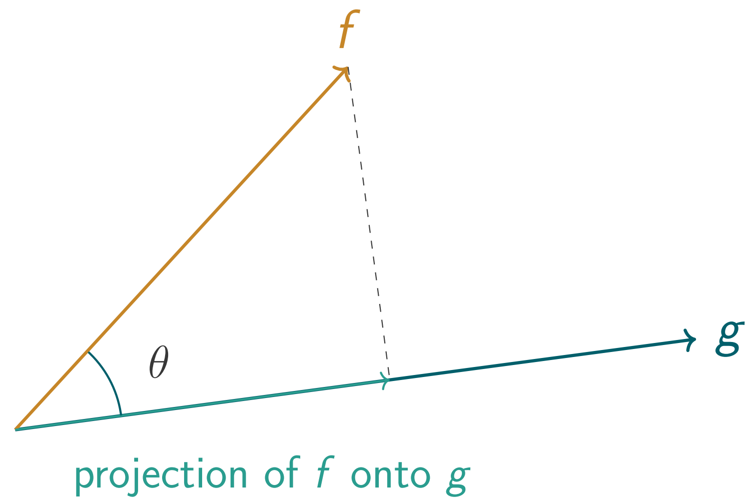
$$\left(\sum_{i=1}^{n} a_i b_i\right)^2 \le \left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right).$$

Continuous form. For continuous functions f, g on $[a, b]$,

$$\left(\int_a^b f(x)g(x)\,dx\right)^2 \le \left(\int_a^b f(x)^2\,dx\right)\left(\int_a^b g(x)^2\,dx\right).$$

Equality holds precisely when f and g are proportional.

Geometric Picture: Cauchy–Schwarz



The integral $\int fg$ measures how much f aligns with g: largest when they rise and fall together, zero when perpendicular.

Cauchy–Schwarz — Example 1, Problem

Problem

Prove that $\int_0^1 \sqrt{x}\, e^{-x}\, dx \;\leq\; \dfrac{\sqrt{1-e^{-2}}}{2}$.

The integrand is already a product of two functions, so Cauchy–Schwarz is the natural tool. The choice $f(x) = \sqrt{x}$ and $g(x) = e^{-x}$ is forced: squaring each collapses the square root and doubles the exponent into manageable integrals.

Cauchy–Schwarz — Example 1, Proof

Apply Cauchy–Schwarz with $f(x) = \sqrt{x}$ and $g(x) = e^{-x}$:

$$\left(\int_0^1 \sqrt{x}\, e^{-x}\, dx\right)^2 \;\le\; \left(\int_0^1 x\, dx\right)\left(\int_0^1 e^{-2x}\, dx\right).$$

Evaluate each integral on the right:

$$\int_0^1 x\, dx = \frac{1}{2}, \qquad \int_0^1 e^{-2x}\, dx = \frac{1-e^{-2}}{2}.$$

Their product is $\dfrac{1-e^{-2}}{4}$. Taking the positive square root,

$$\int_0^1 \sqrt{x}\, e^{-x}\, dx \;\le\; \frac{\sqrt{1-e^{-2}}}{2}. \quad \square$$

Cauchy–Schwarz — Example 2, Problem

Problem

Show that for any continuous function f on $[0,1]$,

$$\left(\int_0^1 f(x)\,dx\right)^2 \;\leq\; \int_0^1 f(x)^2\,dx.$$

This is the continuous analogue of the root-mean-square vs arithmetic-mean inequality. The trick is to pair f with the constant function 1, whose own square integrates to the length of the interval.

Cauchy–Schwarz — Example 2, Proof

Take $g(x) = 1$ in the Cauchy–Schwarz inequality. Then

$$\left(\int_0^1 f(x) \cdot 1 \, dx\right)^2 \le \left(\int_0^1 f(x)^2 \, dx\right)\left(\int_0^1 1^2 \, dx\right).$$

The second integral on the right equals 1, so the inequality simplifies to

$$\left(\int_0^1 f(x) \, dx\right)^2 \le \int_0^1 f(x)^2 \, dx. \quad \square$$

Cauchy–Schwarz — Example 3, Problem

Problem

Let f be a positive continuous function on $[0,1]$. Prove that

$$\left(\int_0^1 f(x)\,dx\right)\left(\int_0^1 \frac{1}{f(x)}\,dx\right) \;\geq\; 1.$$

The left-hand side has exactly the shape of the right-hand side of Cauchy–Schwarz if we can write f and $1/f$ as squares. Since $f > 0$ this is possible: take $\sqrt{f}$ and $1/\sqrt{f}$ as the underlying factors.

Cauchy–Schwarz — Example 3, Proof

Set $u(x) = \sqrt{f(x)}$ and $v(x) = 1/\sqrt{f(x)}$, both continuous because $f > 0$. Then

$$u(x)^2 = f(x), \qquad v(x)^2 = \frac{1}{f(x)}, \qquad u(x)v(x) = 1.$$

Cauchy–Schwarz applied to u and v gives

$$\left(\int_0^1 u(x)v(x)\,dx\right)^2 \;\le\; \left(\int_0^1 u(x)^2\,dx\right)\left(\int_0^1 v(x)^2\,dx\right),$$

which reads

$$1 \;=\; \left(\int_0^1 1\,dx\right)^2 \;\le\; \left(\int_0^1 f(x)\,dx\right)\left(\int_0^1 \frac{1}{f(x)}\,dx\right). \quad \square$$

Projection onto a Polynomial

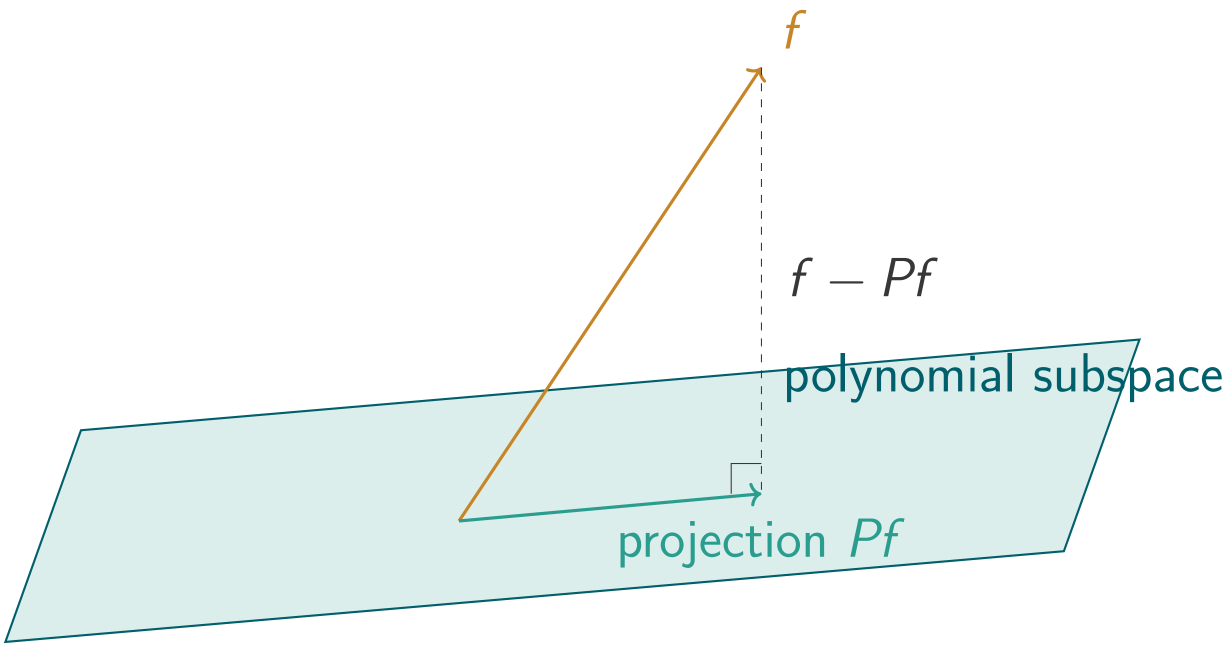
Method

To bound $\int f(x)^2\,dx$ under integral constraints on f, pick a polynomial P that encodes those constraints and apply Cauchy–Schwarz to the pair (f, P).

The constraint $\int f = 0$ means f is orthogonal to constants, so constants yield no information; one then turns to the next orthogonal polynomial $P(x) = x - \frac{1}{2}$, and so on.

Each such P extracts a tighter bound by projecting f onto a richer subspace.

Geometric Picture: Orthogonal Projection



f splits into its projection onto the polynomial subspace plus a remainder perpendicular to that subspace.

Projection — Example 1, Problem

Problem

Find the minimum of $\int_0^1 f(x)^2\,dx$ subject to $\int_0^1 f(x)\,dx = 1$.

The constraint fixes a single linear functional of f, namely its integral. A natural guess is that the minimiser is the simplest function satisfying the constraint — the constant $f \equiv 1$. Cauchy–Schwarz, with $P(x) = 1$ as the projection target, converts this guess into a proof.

Projection — Example 1, Proof

Project onto $P(x) = 1$. Cauchy–Schwarz gives

$$\left(\int_0^1 f(x) \cdot 1 \, dx\right)^2 \le \left(\int_0^1 f(x)^2 \, dx\right)\left(\int_0^1 1 \, dx\right).$$

The right-hand factor is 1, and the left-hand side equals 1 by the constraint, so

$$1 \le \int_0^1 f(x)^2 \, dx.$$

Equality holds iff f is proportional to $P(x) = 1$, i.e. $f \equiv 1$. Hence the minimum is 1, achieved by $f \equiv 1$. $\square$

Projection — Example 2, Problem

Problem

Suppose $\int_0^1 f(x)\,dx = 0$. Prove that

$$\int_0^1 f(x)^2\,dx \;\geq\; 12\left(\int_0^1 xf(x)\,dx\right)^2.$$

The constraint says f is orthogonal to the constant 1, so Example 1's bound collapses. To extract a bound now, we project f onto a linear polynomial that is also orthogonal to 1 — the next element of the Legendre hierarchy on $[0,1]$.

Projection — Example 2, Proof

Seek $P(x) = x - c$ with $\int_0^1 (x - c)\, dx = 0$:

$$\frac{1}{2} - c = 0 \implies c = \frac{1}{2}, \qquad P(x) = x - \tfrac{1}{2}.$$

Because $\int_0^1 f = 0$,

$$\int_0^1 \left(x - \tfrac{1}{2}\right) f(x)\, dx \;=\; \int_0^1 x f(x)\, dx.$$

Cauchy–Schwarz applied to f and P gives

$$\left(\int_0^1 x f(x)\, dx\right)^2 \;\le\; \left(\int_0^1 f(x)^2\, dx\right)\left(\int_0^1 \left(x - \tfrac{1}{2}\right)^2\, dx\right) = \frac{1}{12}\int_0^1 f(x)^2\, dx.$$

Rearranging yields the claim. $\square$

Chebyshev's Integral Inequality

Theorem

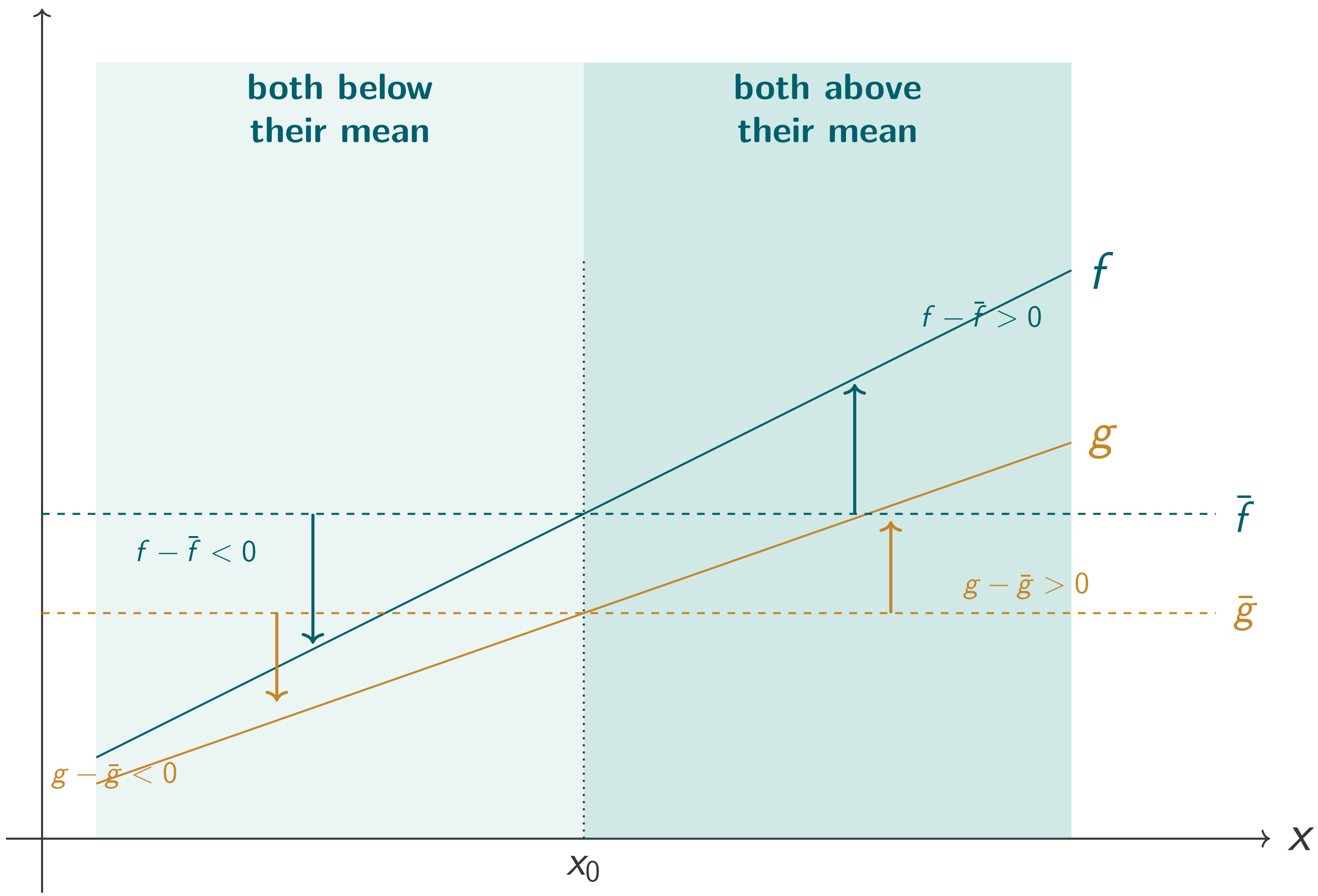
If f and g are both increasing (or both decreasing) on $[a, b]$, then

$$\frac{1}{b-a}\int_a^b f(x)g(x)\,dx \;\geq\; \left(\frac{1}{b-a}\int_a^b f\,dx\right)\left(\frac{1}{b-a}\int_a^b g\,dx\right).$$

If one is increasing and the other decreasing, the inequality reverses.

In words: when two functions rise together, the average of their product exceeds the product of their averages.

Geometric Picture: Chebyshev



At every x the deviations $f(x) - \bar{f}$ and $g(x) - \bar{g}$ share a sign, so their product $(f - \bar{f})(g - \bar{g})$ is ≥ 0 throughout. Averaging over $[a, b]$ gives $\overline{fg} - \bar{f}\bar{g} \geq 0$, which is Chebyshev.

Chebyshev — Example, Problem

Problem

Prove that for every $n \in \mathbb{N}$,

$$\int_0^1 x^n e^x \, dx \;\geq\; \frac{e-1}{n+1}.$$

The integrand is a product of x^n and e^x, both increasing on $[0, 1]$. This is exactly the setting of Chebyshev, which separates the product integral into a product of individual integrals, each elementary.

Chebyshev — Example, Proof

Let $f(x) = x^n$ and $g(x) = e^x$. Both are strictly increasing on $[0,1]$, so Chebyshev's inequality with $b - a = 1$ gives

$$\int_0^1 x^n e^x\,dx \;\geq\; \Big(\int_0^1 x^n\,dx\Big)\Big(\int_0^1 e^x\,dx\Big).$$

Evaluating each factor,

$$\int_0^1 x^n\,dx = \frac{1}{n+1}, \qquad \int_0^1 e^x\,dx = e - 1,$$

hence

$$\int_0^1 x^n e^x\,dx \;\geq\; \frac{e-1}{n+1}. \quad \square$$

Hermite–Hadamard Inequality

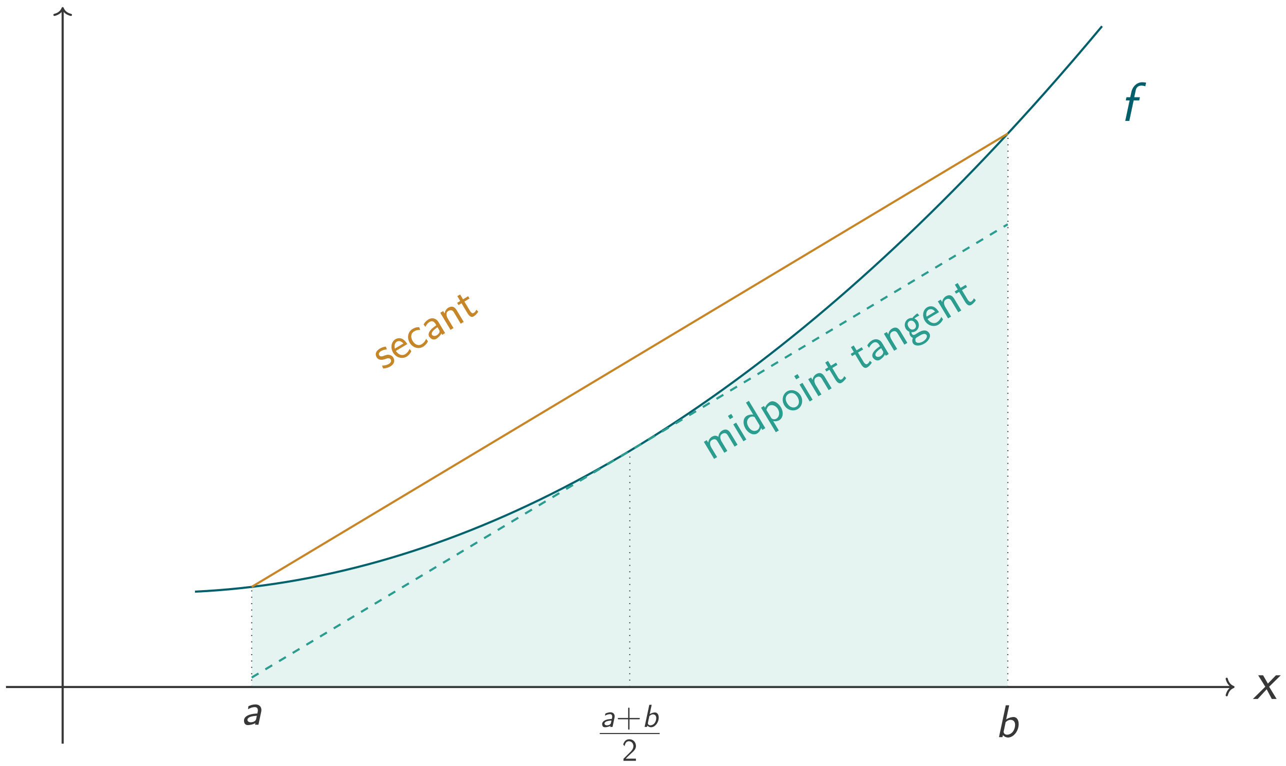
Theorem

If $f : [a, b] \to \mathbb{R}$ is twice differentiable and convex (i.e. $f'' > 0$), then

$$f\left(\frac{a+b}{2}\right) \le \frac{1}{b-a}\int_a^b f(x)\,dx \le \frac{f(a)+f(b)}{2}.$$

The mean value of a convex function lies between its value at the midpoint and the average of its endpoint values.

Geometric Picture: Hermite–Hadamard



The integral lies above the midpoint-tangent trapezoid and below the secant-chord trapezoid.

Hermite–Hadamard — Example, Problem

Problem

Prove that for $a < b$,

$$e^{(a+b)/2} \leq \frac{e^b - e^a}{b - a} \leq \frac{e^a + e^b}{2}.$$

All three quantities have the shape of the three terms in Hermite–Hadamard: midpoint value, mean integral value, and endpoint average. The natural candidate is $f(x) = e^x$, whose convexity on every interval is immediate from $f''(x) = e^x > 0$.

Hermite–Hadamard — Example, Proof

Let $f(x) = e^x$. Since $f''(x) = e^x > 0$, f is strictly convex on every interval, so Hermite–Hadamard gives

$$f\left(\tfrac{a+b}{2}\right) \;\le\; \frac{1}{b-a}\int_a^b f(x)\,dx \;\le\; \frac{f(a)+f(b)}{2}.$$

The middle integral evaluates to

$$\frac{1}{b-a}\int_a^b e^x\,dx \;=\; \frac{e^b - e^a}{b-a}.$$

Substituting and using $f\left(\frac{a+b}{2}\right) = e^{(a+b)/2}$ and $\frac{f(a)+f(b)}{2} = \frac{e^a+e^b}{2}$ yields the claim. $\square$

Practice Problems — Set 1

Problem 1

Let f be continuous on $[0,1]$ with $0 \leq f(x) \leq 1$. Prove that

$$\left(\int_0^1 f(x)\,dx\right)^2 \;\leq\; 2\int_0^1 xf(x)\,dx.$$

Problem 2

Let f be a strictly increasing, positive function on $[0,1]$. Prove that

$$\int_0^1 xf(x)\,dx \;\geq\; \frac{1}{2}\int_0^1 f(x)\,dx.$$

Practice Problems — Set 2

Problem 3

Prove that for every continuous f on $[0, \pi]$,

$$\int_0^\pi f(x)\sin(x)\,dx \;\le\; \int_0^\pi f(x)^2\,dx + \frac{\pi}{8}.$$

Problem 4

Using Hermite–Hadamard, prove that

$$2 \;<\; \int_1^3 x\ln(x)\,dx \;<\; 3\ln(3).$$

Practice Problems — Set 3

Problem 5

Prove that

$$\int_0^{\pi/2} \sqrt{\sin x}\, dx \;\le\; \sqrt{\frac{\pi}{2}}.$$

Problem 6

If f' is continuous on $[0,1]$ and $f(0) = 0$, prove the Poincaré-type inequality

$$\int_0^1 f(x)^2\, dx \;\le\; \frac{1}{2}\int_0^1 f'(x)^2\, dx.$$

Solution to Problem 2

Recap. f is increasing and positive on $[0, 1]$; show $\int_0^1 xf(x)\,dx \geq \frac{1}{2}\int_0^1 f(x)\,dx$.

Proof. Set $g(x) = x$. Both f and g are increasing on $[0, 1]$, so Chebyshev's inequality (with $b - a = 1$) gives

$$\int_0^1 xf(x)\,dx \;\geq\; \Big(\int_0^1 f(x)\,dx\Big)\Big(\int_0^1 x\,dx\Big) \;=\; \frac{1}{2}\int_0^1 f(x)\,dx. \qquad \square$$

Solution to Problem 3 (Setup)

Recap. Prove $\int_0^\pi f(x)\sin(x)\,dx \le \int_0^\pi f(x)^2\,dx + \frac{\pi}{8}$.

Proof. The shape of the target inequality suggests a completed square. Starting from the non-negative integrand

$$\int_0^\pi \left(f(x) - \tfrac{1}{2}\sin(x)\right)^2 dx \;\ge\; 0,$$

expand:

$$\int_0^\pi f(x)^2\,dx \;-\; \int_0^\pi f(x)\sin(x)\,dx \;+\; \frac{1}{4}\int_0^\pi \sin^2(x)\,dx \;\ge\; 0.$$

Solution to Problem 3 (Finish)

Rearranging,

$$\int_0^\pi f(x)\sin(x)\,dx \;\le\; \int_0^\pi f(x)^2\,dx \;+\; \frac{1}{4}\int_0^\pi \sin^2(x)\,dx.$$

The remaining integral is standard, using $\sin^2(x) = \frac{1-\cos(2x)}{2}$:

$$\int_0^\pi \sin^2(x)\,dx \;=\; \left[\frac{x}{2} - \frac{\sin(2x)}{4}\right]_0^\pi \;=\; \frac{\pi}{2}.$$

Substituting back,

$$\int_0^\pi f(x)\sin(x)\,dx \;\le\; \int_0^\pi f(x)^2\,dx \;+\; \frac{\pi}{8}. \quad \square$$

Thank you

Questions?